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DOC REF: AREL-OPS-2026-OPHR-V1 | **TECHNICAL WHITE PAPER & OPERATING DYNAMICS GUIDE**
Subject: Solar PV vs. Hybrid BESS Daily Operating Hours in Bangladesh • Published: November 2026

Solar vs. Hybrid Daily Operating Hours & Power Generation Dynamics

An engineering and commercial evaluation of daily generation windows, Peak Sun Hours (PSH), anti-islanding grid protocols, and 24-hour hybrid battery autonomy under Bangladesh climate and utility conditions.

GRID-TIED GENERATION WINDOW	BANGLADESH ANNUAL MEAN PSH	HYBRID + BESS AVAILABILITY
7.5 – 8.5 Hrs	**3.84 PSH**	**24/7 Continuous**
08:00 AM – 04:30 PM Daytime	1,400 kWh / kWp / year yield	<10ms Industrial Online UPS

1. THE TWO ENGINEERING DEFINITIONS OF "OPERATING HOURS"

When commercial clients in Bangladesh ask *"What is the daily operating hour?"*, electrical engineering distinguishes between two fundamental concepts:

- **1. Active Daylight Generation Window (Clock Hours: 7.5 to 8.5 Hours/Day)**: The duration from sunrise to sunset when solar radiation is high enough (>150 W/m²) for inverters to synchronize with the AC busbar. In Bangladesh (23.8°N latitude), this spans from approximately **08:00 AM to 04:30 PM**.
- **2. Peak Sun Hours (PSH / Full-Load Equivalent: 3.84 to 4.20 Hours/Day)**: Because solar radiation varies in an arc (bell curve), solar engineers convert fluctuating irradiance into the equivalent number of hours at standard peak test intensity (1,000 W/m²). In Bangladesh, 1 kWp of installed N-Type TOPCon bifacial modules generates an annual average of **3.84 to 4.20 kWh per day**.

2. MASTER ARCHITECTURE COMPARISON: GRID-TIED VS. HYBRID BESS

ENGINEERING PARAMETER	STANDARD GRID-TIED SOLAR (NO BATTERY)	HYBRID SOLAR + BESS (LIFEPO4 STORAGE)
Daily Operating Window	**7.5 to 8.5 Hours / Day** (Sunlight only)	**Up to 24 Hours / Day** (Continuous 24/7)
Daily Full-Load Yield (PSH)	**3.84 to 4.20 Peak Sun Hours / Day**	**3.84 PSH Daytime + 4 to 12h Battery Autonomy**
Peak Production Period	**11:00 AM – 02:00 PM** (85%–100% capacity)	**11:00 AM – 02:00 PM** (Direct Solar + Battery Rapid Charge)
Utility Grid Outage Behavior	**Automatic Shutdown in <20ms** (SREDA anti-islanding)	**Zero Interruption (<10ms Seamless UPS transfer)**
Evening / Night Availability	**0 Hours** (Factory purchases grid or diesel power)	**4 to 12 Hours** (Discharges stored clean daytime solar)
Evening Peak Tariff Shaving	None (Draws expensive grid peak at Tk 17.50–19.80)	**100% Shaved** (Saves Tk 10.00–12.00/kWh net arbitrage)
Diesel Generator (DG) Fuel Replacement	Displaces daytime DG only if sync-controller installed	**Replaces 100% of Daytime & Evening Diesel Fuel**
Levelized Cost of Energy (LCOE)	**Tk 3.80 – 4.50 / kWh** (Lowest CapEx, 3.0-yr payback)	**Tk 7.50 – 9.20 / kWh** (50% cheaper than grid peak)

Key Takeaway: Grid-Tied Solar is the champion for **daytime cost reduction** (8 AM – 5 PM shifts), whereas Hybrid Solar with BESS is the sovereign solution for **uninterrupted 24/7 manufacturing** and zero diesel reliance.

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3. THE 24-HOUR HOUR-BY-HOUR OPERATIONAL DISPATCH CURVE

Illustrated below is the dispatch logic for a manufacturing facility across a complete 24-hour cycle under both system topologies:

TIME WINDOW	SOLAR RADIATION	STANDARD GRID-TIED SOLAR DISPATCH	HYBRID SOLAR + LIFEPO4 BESS DISPATCH
06:00 – 08:00	Low (50–250 W/m²)	Inverter wakes up. Offsets 10%–25% of morning machinery startup load.	Solar powers critical loads. Inverter assesses battery state-of-charge (SoC).
08:00 – 11:00	**Medium-High (300–700 W/m²)**	**Rapid output ramp. Solar delivers 50%–75% of total plant electricity.**	**Solar directly runs factory machinery. Excess power charges battery bank.**
11:00 – 14:00	**Peak (800–1,050 W/m²)**	**Solar Noon Peak (85%–100%).** Excess weekend power exported at Tk 10.50.	**Full Factory Load Powered** + Battery reaches 100% full charge at 1C/0.5C rate.
14:00 – 16:30	Declining (650–200 W/m²)	Solar tapers from 70% to 25%. Utility grid smoothly supplements the balance.	Solar continues daytime support. Battery sits fully charged on standby for peak.
16:30 – 18:00	**Dusk / Standby (<100 W/m²)**	**Inverters power down into night standby mode. 100% load transfers to grid.**	**Seamless Evening Transition:** Battery engages instantly as sun sets.
18:00 – 23:00	**Zero (Night)**	**Zero Generation:** Factory purchases expensive evening peak grid tariff.	**Peak Shaving:** Battery supplies plant during high-tariff period (Tk 17.50+).
23:00 – 06:00	Zero (Night)	Facility runs on off-peak grid power or diesel generators during outages.	Supplies night shifts or recharges at low off-peak grid rate if needed for next day.

4. BATTERY AUTONOMY OPTIONS FOR HYBRID SYSTEMS

2-Hour Peak Shaving Pack (C/2 Rate)

Operating Window: 17:30 – 19:30 (Evening Peak).
Objective: Eliminates peak tariff penalties (Tk 17.50–19.80/kWh) and buffers brief momentary grid voltage sags. Lowest storage CapEx.

4-Hour Evening Shift Pack (C/4 Rate)

Operating Window: 17:00 – 21:00 (Full 2nd Shift).
Objective: Powers garment sewing lines and packaging during the entire evening shift without relying on noisy, polluting diesel gensets.

8-Hour Full Night Shift Pack (C/8 Rate)

Operating Window: 17:00 – 01:00 (Night Operation).
Objective: Complete replacement of night diesel generation. Eliminates hundreds of liters of diesel fuel consumption every evening.

24/7 Deep Industrial Autonomy (12–16h Pack)

Operating Window: Continuous 24/7/365 Non-Stop.
Objective: Critical continuous-flow industries (Spinning mills, Pharmaceutical cleanrooms, Cold chain food processing). 100% islanded capability.

5. CRUCIAL GRID COMPLIANCE: WHY STANDARD SOLAR SHUTS DOWN IN OUTAGES

Statutory Anti-Islanding Rule (IEEE 1547 / SREDA): When BREB, BPDB, or DESCO grid voltage drops to zero during load-shedding, standard grid-tied string inverters **must disconnect within 20 milliseconds**. This prevents dangerous back-feeding of solar power into utility poles, protecting utility repair crews.

The Hybrid Solution: Hybrid inverters incorporate an automated internal AC disconnect switch. In an outage, they disconnect the grid within **<10 milliseconds** while continuing to power the factory from solar and batteries as an independent microgrid.

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6. BANGLADESH 12-MONTH SEASONAL OPERATING HOUR & YIELD MATRIX

Empirical solar irradiance data (NREL / Bangladesh Meteorological Department) across the 4 distinct climate seasons:

MONTH	SEASON CATEGORY	DAYLIGHT WINDOW	DAILY PEAK SUN HOURS (PSH)	SPECIFIC YIELD (KWH/KWP)	ATMOSPHERIC CONDITIONS & TECHNICAL NOTES
January	Winter	06:35 – 17:35 (11.0h)	**3.92 PSH**	121.5 kWh	Clear skies; cold ambient air (+4% thermal panel efficiency boost).
February	Winter / Spring	06:25 – 17:55 (11.5h)	**4.35 PSH**	121.8 kWh	Rapidly rising insolation; high clear-sky clarity across central plains.
March	**Summer (Pre-Monsoon)**	**06:05 – 18:10 (12.1h)**	**4.85 PSH**	**150.4 kWh**	**Strong direct solar radiation; optimal tilt yields peak generation.**
April	**Summer (Peak Sun)**	**05:40 – 18:25 (12.8h)**	**5.12 PSH**	**153.6 kWh**	**Annual Solar Peak: Maximum irradiation across Bangladesh.**
May	**Pre-Monsoon / Summer**	**05:20 – 18:40 (13.3h)**	**4.72 PSH**	**146.3 kWh**	**Extended daylight span; intermittent pre-monsoon convective rains.**
June	**Monsoon Season**	**05:15 – 18:50 (13.6h)**	**3.42 PSH**	**102.6 kWh**	**Monsoon cloud cover; N-Type TOPCon bifacials harvest diffuse light.**
July	**Monsoon Season**	**05:22 – 18:48 (13.4h)**	**3.20 PSH**	**99.2 kWh**	**Heavy precipitation; natural rainwater automatically cleans modules.**
August	**Monsoon Season**	**05:32 – 18:35 (13.1h)**	**3.52 PSH**	**109.1 kWh**	**Breaks in cloud cover; rapid generation spikes during sun windows.**
September	Post-Monsoon	05:42 – 18:10 (12.5h)	**3.95 PSH**	118.5 kWh	Clearing atmosphere; steady autumn generation profile.
October	Autumn	05:52 – 17:40 (11.8h)	**4.25 PSH**	131.8 kWh	Exceptional atmospheric clarity; steady and predictable power curves.
November	Late Autumn / Winter	06:08 – 17:20 (11.2h)	**4.10 PSH**	123.0 kWh	Mild winter conditions; dry season commences.
December	Winter Solstice	06:28 – 17:18 (10.8h)	**3.82 PSH**	118.4 kWh	Shortest daylight of the year; morning mist burns off by 08:30 AM.
ANNUAL TOTAL / MEAN			**3.84 PSH / Day**	**1,400 kWh / kWp**	**Bankable 25-Year Specific Yield Basis for Bangladesh**

7. FINANCIAL ECONOMICS: DIESEL GENERATOR FUEL REPLACEMENT

During utility grid blackouts, industrial enterprises in Gazipur, Savar, and Narayanganj rely on diesel generators. The comparative cost dynamics are stark:

Diesel Generator Running Cost
Tk 32.00 – 38.00 / kWh
1 Liter Diesel (~Tk 107) yields ~3.0–3.2 kWh electric power + lube & maintenance

Ariel Hybrid Stored Solar Cost
Tk 7.50 – 9.20 / kWh
Levelized cost of solar PV + 6,000-cycle CATL/BYD LiFePO4 battery pack

Net Direct Fuel Savings: Tk 24.50 to 29.00 Saved Per Kilowatt-Hour of Load-Shedding Displaced!

8. TECHNICAL RECOMMENDATION MATRIX FOR CLIENTS

FACTORY SHIFT STRUCTURE	RECOMMENDED SYSTEM	DAILY OPERATING WINDOW	PRIMARY FINANCIAL BENEFIT
1 Single Shift (08:00 – 17:00)	**Standard Grid-Tied Solar**	**7.5 – 8.5 Hours/Day** (Daylight)	Fastest payback (3.0 years); lowest capital expenditure.
2 Shifts (08:00 – 22:00)	**Hybrid Solar + 4h Battery**	**13.5 Hours/Day** (Day + Evening)	Shaves expensive peak tariff; seamless shift continuity.
Continuous 24/7 Production	**Hybrid Solar + 8h–12h Battery**	**24 Hours/Day Non-Stop**	Zero diesel burn; 100% immunity from national load-shedding.